

Date: _____

Day: _____

Simple Linear Regression:

$$Y = \beta_0 + \beta_1 x$$

$\downarrow$ $\downarrow$
intercept slope

Our data doesn't follow this line, so the following formula represents points along the line.

$$Y = \beta_0 + \beta_1 x + \varepsilon$$

- x is independent variable / regressor (random)
- β_0, β_1 are unknown constants.
- ε (epsilon indicates randomness)

Expected values and variance.

$$\Rightarrow E[\varepsilon] = 0, \text{Var}[\varepsilon] = \sigma^2$$

$$\Rightarrow E[Y/x] = E[\beta_0 + \beta_1 x + \varepsilon]$$

$$E[Y/x] = \beta_0 + \beta_1 x$$

True regression line along which the data is represented.

$$\Rightarrow \text{Var}[Y/x] = \text{Var}[\beta_0 + \beta_1 x + \varepsilon] = \sigma^2$$

Non-Random variable has $\text{var}=0$.

Date: _____

Day: _____

b_0, b_1 are the estimates of β_0, β_1

So,

$$\hat{y} = b_0 + b_1 x$$

predicted y

⇒ Prediction Error:

$$e_i = y_i - \hat{y}_i$$

Actual - Predicted.

$$y_i - (b_0 + b_1 x_i)$$

If ϵ follows Gaussian distribution then the peak of every of curve is on true regression line.

⇒ Sum of Squared Errors:

$$SSE = \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

$$= \sum_{i=1}^N (y_i - b_0 - b_1 x_i)^2$$

To maximize, take derivative

To minimize, take 2nd derivative.

Date: _____

Day: _____

$$\frac{\partial SSE}{\partial b_0}, \quad \frac{\partial SSE}{\partial b_1}$$

we have to minimize this.

$$\Rightarrow \frac{\partial SSE}{\partial b_0} = -2 \sum_{i=1}^N (y_i - b_0 - b_1 x_i)$$

$$\Rightarrow \frac{\partial SSE}{\partial b_1} = -2 \sum_{i=1}^N (y_i - b_0 - b_1 x_i) x_i$$

• Setting partial derivatives equal to zero and rearranging terms we obtain.

$$\Rightarrow \frac{\partial SSE}{\partial b_0} = n b_0 + b_1 \sum_{i=1}^N x_i = \sum_{i=1}^N y_i$$

$$\Rightarrow \frac{\partial SSE}{\partial b_1} = b_0 \sum_{i=1}^N x_i + b_1 \sum_{i=1}^N x_i^2 = \sum_{i=1}^N x_i y_i$$

These equations can be solved simultaneously to compute b_0 , b_1 .

Date: _____

Day: _____

$$\left[\begin{array}{l} b_0 = \frac{\sum_{i=1}^n y_i - b_1 \sum_{i=1}^n x_i}{n} \\ \text{OR} \\ b_0 = \bar{y} - b_1 \bar{x} \end{array} \right]$$

$$\left[\begin{array}{l} b_1 = \frac{n \sum_{i=1}^N x_i y_i - (\sum_{i=1}^N x_i) (\sum_{i=1}^N y_i)}{n \sum_{i=1}^N x_i^2 - (\sum_{i=1}^N x_i)^2} \end{array} \right]$$

Question: Predicting House Prices using Linear Regression

A real estate company wants to predict house prices based on house size (sq. ft.). The given dataset is:

House Size ($\bar{X}$) (sq. ft.)	Price ($\bar{Y}$) (\$1000s)
1400	245
1600	312
1700	279
1875	308
1100	199

A simple linear regression model is estimated as:

$$\hat{Y} = 50 + 0.12X$$

- Interpret the coefficients:**
 - What does the intercept (50) represent?
 - What does the slope (0.12) represent?
- Make predictions** using the given regression equation:
- Calculate the residuals** (difference between actual and predicted prices) for each house.
- Analyze the model:**
 - If a house's actual price is much higher or lower than the predicted price, what could be the reason?
 - What other factors might improve the accuracy of the model?

Part 1:

=> Intercept (50):

if a house had 0 sq.ft., the predicted price would be \$50,000.

=> Slope (0.12):

For every 1 additional sq.ft., the predicted price is increased by $0.12 \times 1000 = \$120$.

Part 2:

Predicted $\hat{y}$

• For 1400 sq.ft

$$\begin{aligned}\hat{y} &= 50 + 0.12(1400) \\ &= 218 \\ &= 218 \times 1000 = \$218,000\end{aligned}$$

• For 1600 sq.ft

$$\begin{aligned}\hat{y} &= 50 + 0.12(1600) \\ &= 242 \times 1000 \\ &= 242,000\end{aligned}$$

• For 1700

$$\begin{aligned}\hat{y} &= 254 \times 1000 \\ &= 254,000\end{aligned}$$

• For 1875

$$\begin{aligned}\hat{y} &= 275 \times 1000 \\ &= 275,000\end{aligned}$$

• For 1100

$$\begin{aligned}\hat{y} &= 182 \times 1000 \\ &= 182,000.\end{aligned}$$

Date: _____

Day: _____

Part 3:

Residuals = $Y_{\text{actual}} - Y_{\text{predicted}}$.

X	Y	$\hat{Y}$	Residual
1400	245	218	27
1600	312	242	70
1700	279	254	25
1875	308	275	33
1100	199	182	17

Part 4:

⇒ The model estimates house price based on size, but residuals show deviations due to other factors (e.g., location, condition)

⇒ The model might need improvements by including more variables like ^{no of} bedrooms, neighbourhood etc to increase accuracy.

Question 1: A business owner wants to analyze how temperature affects ice cream sales. The following data has been collected:

Temperature (X) ($^{\circ}\text{C}$)	Ice Cream Sales (Y) (cones)
25	200
30	300
35	400

Determine the regression equation ,Predict sales for new temperatures:

- Estimate ice cream sales when the temperature is 28°C . And 40°C .

Date: _____ Day: _____

1. $\hat{Y} = b_0 + b_1 X$

$$b_0 = \bar{Y} - b_1 \bar{X}$$

$$= 300 - (20)(30)$$

$$b_1 = \frac{n \sum xy - (\sum x)(\sum y)}{n \sum x^2 - (\sum x)^2}$$

$$= \frac{(3)(28000) - (90)(900)}{(3)(2750) - (90)^2}$$

$$b_1 = 20$$

$$b_0 = -300$$

$$\boxed{\hat{Y} = -300 + 20X}$$

2. Predictions:

Sales at 28°C

$$\hat{Y} = -300 + 20(28) = 260$$

Sales at 40°C

$$\hat{Y} = -300 + 20(40) = 500$$

Question 3: Compute SSE (Sum of Squared Errors) and MSE (Mean Squared Error)

Given the **actual vs. predicted values**:

Actual Y	Predicted $\hat{Y}$
10	12
15	14
20	18
25	22

Dates: _____ *Page:* _____

$$\begin{aligned}\Rightarrow \text{SSE} &= \sum (Y - \hat{Y})^2 \\ &= (10 - 12)^2 + (15 - 14)^2 + (20 - 18)^2 \\ &\quad + (25 - 22)^2 \\ &= 4 + 1 + 4 + 9 \\ &= 18\end{aligned}$$
$$\Rightarrow \text{MSE} = \frac{\text{SSE}}{n-2} = \frac{18}{4-2} = 9.$$

Question:

Temperature (X)	Converted Sugar
1.0	8.1
1.1	7.8
1.2	8.5
1.3	9.8
1.4	9.5
1.5	8.9
1.6	8.6
1.7	10.2
1.8	9.3
1.9	9.2
2.0	10.5

$$Y = b_0 + b_1 X$$

$$b_1 = \frac{n \sum xy - (\sum x)(\sum y)}{n \sum x^2 - (\sum x)^2}$$

$$= \frac{(11)(152.59) - (16.5)(100.4)}{(11)(25.85) - (272.25)}$$

$$b_1 = 1.81$$

$$\begin{aligned} b_0 &= \bar{y} - b_1 \bar{x} \\ &= (9.127) - (1.81)(1.5) \\ &= 6.412 \end{aligned}$$

$$Y = 6.412 + (1.81) X$$